



WATER APPLICATION VALUE ENGINE

WATER SOLUTIONS

WAVE Program Version: 1.85.29

Calculation Engine Version: 01.12.40.01

Database Version: 90



Project Name: Wave-RO-4/4/2026 1_v1

Case Name: Case 1

Customer:

Prepared by: Aryan

Company: DuPont

Country:

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Project Notes:

Case #: 1 of: 1

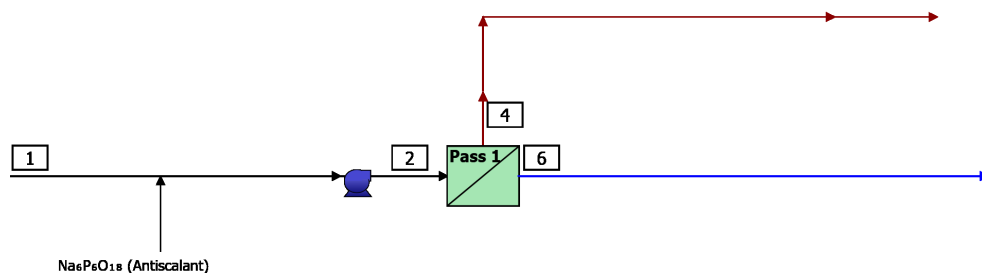
Case Notes: Case 1

Keywords:

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RO Detailed Report

RO System Flow Diagram



#	Description	Flow (m ³ /h)	TDS (mg/L)	Pressure (bar)
1	Raw Feed to RO System	85.3	602.3	0.0
2	Net Feed to Pass 1	85.3	602.4	3.7
4	Total Concentrate from Pass 1	17.1	2,854	2.4
6	Net Product from RO System	68.2	36.67	0.0

RO System Overview

Total # of Units	1	Online =	1	Standby =	0	RO Recovery	80.0 %
System Flow Rate	(m ³ /h)	Net Feed =	85.3	Net Product =	68.2		

Pass	Pass 1
Stream Name	Stream 1
Water Type	Surface Water (SDI < 5)
Number of Elements	90
Total Active Area (m ²)	3345
Feed Flow per Pass (m ³ /h)	85.3
Feed TDS ^a (mg/L)	602.4
Feed Pressure (bar)	3.7
Flow Factor Per Stage	0.85, 0.85
Permeate Flow per Pass (m ³ /h)	68.2
Pass Average flux (LMH)	20.4
Permeate TDS ^a (mg/L)	36.67
Pass Recovery	80.0 %
Average NDP (bar)	2.2
Specific Energy (kWh/m ³)	0.16
Temperature (°C)	25.0
pH	7.0 (After Adjustment)
Chemical Dose	3.0 mg/L Na ₆ P ₆ O ₁₈
RO System Recovery	80.0 %
Net RO System Recovery	80.0%

Footnotes:

^aTotal Dissolved Solids and Conductivity includes ions, SiO₂ and B. It does not include NH₃ and CO₂

RO Flow Table (Stage Level) - Pass 1

Stage	Elements	#PV	#Els per PV	Feed				Concentrate			Permeate			
				Feed Flow	Recirc Flow	Feed Press	Boost Press	Conc Flow	Conc Press	Press Drop	Perm Flow	Avg Flux	Perm Press	Perm TDS
				(m ³ /h)	(m ³ /h)	(bar)	(bar)	(m ³ /h)	(bar)	(bar)	(m ³ /h)	(LMH)	(bar)	(mg/L)
1	NF90-400/34i	10	6	85.3	0.00	3.4	0.0	32.2	3.0	0.5	53.1	23.8	0.0	24.61
2	NF90-400/34i	5	6	32.2	0.0	2.8	0.0	17.1	2.4	0.4	15.1	13.6	0.0	79.10

RO Solute Concentrations - Pass 1

Concentrations (mg/L as ion)						
	Feed	Concentrate		Permeate		
		Stage1	Stage2	Stage1	Stage2	Total
NH ₄ ⁺	0.00	0.00	0.00	0.00	0.00	0.00
K ⁺	0.00	0.00	0.00	0.00	0.00	0.00
Na ⁺	29.11	70.23	121.3	4.16	12.68	6.05
Mg ⁺²	46.57	121.8	226.7	0.97	3.38	1.51
Ca ⁺²	58.22	151.9	282.4	1.38	4.83	2.14
Sr ⁺²	0.00	0.00	0.00	0.00	0.00	0.00
Ba ⁺²	0.00	0.00	0.00	0.00	0.00	0.00
CO ₃ ⁻²	0.40	3.12	11.53	0.00	0.00	0.00
HCO ₃ ⁻	429.3	1,113	2,053	12.03	40.85	18.39
NO ₃ ⁻	8.59	17.95	27.55	2.91	7.12	3.84
F ⁻	0.00	0.00	0.00	0.00	0.00	0.00
Cl ⁻	25.77	63.11	110.1	3.11	10.07	4.66
Br ⁻¹	0.00	0.00	0.00	0.00	0.00	0.00
SO ₄ ⁻²	4.29	11.30	21.18	0.05	0.16	0.07
PO ₄ ⁻³	0.00	0.00	0.00	0.00	0.00	0.00
SiO ₂	0.00	0.00	0.00	0.00	0.00	0.00
Boron	0.00	0.00	0.00	0.00	0.00	0.00
CO ₂	54.82	55.98	59.35	54.92	56.80	55.36
TDS ^a	602.3	1,553	2,854	24.61	79.10	36.67
Cond. μS/cm	737	1,784	3,107	37	113	54
pH	7.0	7.4	7.6	5.8	6.2	5.7

Footnotes:

^aTotal Dissolved Solids and Conductivity includes ions, SiO₂ and B. It does not include NH₃ and CO₂

RO Design Warnings

Design Warning	Limit	Value	Pass	Stage	Element	Product
Concentrate Flow Rate < Minimum Limit (m ³ /h)	3.41	3.22	1	1	6	NF90-400/34i
Element Recovery > Maximum Limit (%)	15.0	15.2	1	1	4	NF90-400/34i
Element Recovery > Maximum Limit (%)	15.0	16.9	1	1	5	NF90-400/34i
Element Recovery > Maximum Limit (%)	15.0	19.0	1	1	6	NF90-400/34i

Special Comments

None

RO Flow Table (Element Level) - Pass 1

Stage	Element	Element Name	Recovery (%)	Feed Flow (m³/h)	Feed Press (bar)	Feed TDS (mg/L)	Conc Flow (m³/h)	Perm Flow (m³/h)	Perm Flux (LMH)	Perm TDS (mg/L)
1	1	NF90-400/34i	11.9	8.53	3.4	602.4	7.51	1.02	27.3	14.90
1	2	NF90-400/34i	12.8	7.51	3.3	681.7	6.55	0.96	25.8	17.67
1	3	NF90-400/34i	13.9	6.55	3.2	778.9	5.65	0.91	24.4	21.19
1	4	NF90-400/34i	15.2	5.65	3.1	900.7	4.79	0.86	23.1	25.64
1	5	NF90-400/34i	16.9	4.79	3.1	1,058	3.98	0.81	21.8	32.01
1	6	NF90-400/34i	19.0	3.98	3.0	1,266	3.22	0.76	20.3	41.33
2	1	NF90-400/34i	10.0	6.44	2.8	1,553	5.80	0.64	17.2	50.71
2	2	NF90-400/34i	10.1	5.80	2.7	1,718	5.22	0.58	15.7	60.49
2	3	NF90-400/34i	10.1	5.22	2.6	1,903	4.69	0.53	14.2	72.28
2	4	NF90-400/34i	10.1	4.69	2.5	2,108	4.21	0.48	12.8	86.80
2	5	NF90-400/34i	10.1	4.21	2.5	2,336	3.79	0.42	11.4	104.7
2	6	NF90-400/34i	9.9	3.79	2.4	2,585	3.41	0.38	10.1	127.0

Footnotes:

*Total Dissolved Solids and Conductivity includes ions, SiO₂ and B. It does not include NH₃ and CO₂

RO Solubility Warnings

Warning	Pass No
Langelier Saturation Index > 0	1
Anti-scalants may be required. Consult your anti-scalant manufacturer for dosing and maximum allowable system recovery.	1

RO Chemical Adjustments

	Pass 1 Feed	RO 1 st Pass Conc
pH	7.0	7.6
Langelier Saturation Index	-0.19	1.67
Stiff & Davis Stability Index	0.38	1.66
TDS ^a (mg/l)	602.3	2,854
Ionic Strength (molal)	0.01	0.05
HCO ₃ ⁻ (mg/L)	429.3	2,053
CO ₂ (mg/l)	54.82	59.35
CO ₃ ⁻² (mg/L)	0.40	11.53
CaSO ₄ (% saturation)	0.06	0.78
BaSO ₄ (% saturation)	0.00	0.00
SrSO ₄ (% saturation)	0.00	0.00
CaF ₂ (% saturation)	0.00	0.00
SiO ₂ (% saturation)	0.00	0.00
Mg(OH) ₂ (% saturation)	0.00	0.02

Footnotes:

*Total Dissolved Solids and Conductivity includes ions, SiO₂ and B. It does not include NH₃ and CO₂

RO Utility and Chemical Costs

Service Water

	Flow Rate (m ³ /h)	Unit Cost (\$/m ³)	Hourly Cost (\$/h)	Daily Cost (\$/d)
Non-Product Feed Water				
Pass 1	17.1	0.1400	2.39	57.35
Total Non-product Feed Water Cost	17.1		2.39	57.35
Waste Water Disposal				
Pass 1	17.1	0.6900	11.78	282.65
Total Waste Water Disposal	17.1		11.78	282.65
Total Service Water Cost				340.00

Electricity

Peak Power	(kW)	11.1
Energy	(kWh/d)	266.4
Electricity Unit Cost	(\$/kWh)	0.0900
Electricity Cost	(\$/d)	24.0
Specific Energy	(kWh/m ³)	0.16

Pump	Flow Rate (m ³ /h)	Power (kW)	Energy (kWh/d)	Cost (\$/d)
Pass 1				
Feed	85.29	11.10	266.38	23.97
Pass 1 Total		11.10	266.38	23.97
System Total		11.10	266.38	23.97

Chemical

Chemical	Unit Cost (\$/kg)	Dose (mg/L)	Volume (L/d)	Cost (\$/d)
Na ₆ P ₆ O ₁₈ (100%)(Pass 1)	1.000	3.0	2.5	6.14
Total Chemical Cost				6.1

Utility and Chemical Cost	(\$/d)	370.1
Specific Water Cost	(\$/m ³)	0.226

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